Mario Alejandro *Marroquín*

APPLIED MATHEMATICS · QUANTITATIVE RISK · MACHINE LEARNING

El Salvador
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GitHub: github.com/mariomarroquin3
CryptoRisk: cryptorisk-mauve.vercel.app

Technical high-school student researching financial risk, cryptography and machine learning. Winner of the first National AI Hackathon (2025). I make the design and architecture decisions and build with AI agents.

EXPERIENCE

Accounting assistant · Process automation — Feb 2026 – present

I automate accounting document processing with Python and VBA: extraction, cleaning and validation. Built Partidas DTE, an app that automatically merges the invoices tied to each accounting entry.

Community service · Math teaching — 200 hours

Taught mathematics classes, including to grades above my own.

RESEARCH & PROJECTS

CryptoRisk — Python · PyTorch · Next.js · LaTeX

Out-of-sample comparison of 20+ volatility and tail-risk models (1-day VaR and ES) for BTC and ETH with a 500-day walk-forward protocol and Kupiec, Christoffersen, MCS and Diebold-Mariano tests. Live dashboard.

Entropy Arena — Python · NIST SP 800-22/90B

Framework comparing randomness sources for key generation (BIP39/32/48/67). Recovered MT19937 state in 0.02 s and RANDU in 0.26 s.

Violence, institutions and growth — Python · statsmodels

Reproducible two-way fixed-effects pipeline on 8 countries (2000–2024), with small-cluster robust inference (wild cluster bootstrap, CR2, Driscoll-Kraay) and ML triangulation.

RetinoVisionAI — PyTorch · Flask

Diabetic retinopathy classification with EfficientNet-B3, Grad-CAM interpretability and a web app. Entered in the EUREKA challenge.

Financial data synthesizer (VAE) — PyTorch · Streamlit

Generates synthetic transactions to test accounting software without exposing real data; Streamlit app with PCA validation.

IoT & monitoring — C++ · React · Firebase · Zabbix

RFID access control with ESP32 and Firebase and a real-time dashboard; host monitoring with Zabbix on Ubuntu (team project).

ISO 27001 audit — ISO 27001

Internal audit of a savings and credit cooperative: risks, access controls, policies and a gap report.

ACHIEVEMENTS

- Winner, first National Artificial Intelligence Hackathon (2025), coffee quality classification category.
- Multiple wins in district and departmental public speaking contests.
- Mathematics olympiads, 2 consecutive years.
- Straight-10 grades and first place in class every year.

EDUCATION

Technical Baccalaureate in Technology Infrastructure and IT Services
El Salvador · graduating 2026

SKILLS

Languages
Python, SQL, LaTeX

In progress, AI-assisted
TypeScript, JavaScript/React, Java, VBA, R

ML & data
PyTorch, scikit-learn, statsmodels, arch, Pandas

Applied AI
AI agents, MCP, architecture design

Infrastructure & IoT
Zabbix, Ubuntu, Firebase, ESP32

Areas
Time series, econometrics, cryptography, linear algebra

PROGRAMS & CERTIFICATES

- **Cohere Labs** Open Science Community · ML Summer School 2026
- **Stanford** Probability for AI (admitted, fall 2026) · Code in Place 2026
- **GCI** Matsuo Lab, Univ. of Tokyo (in progress)
- **CUBO+** and **CUBO_ai** (in progress)
- **AWS** ML Engineer Associate (curriculum), Intro to ML, SimuLearn Cloud
- **Cisco** Intro to Data Science · **Udacity** AWS AI Practitioner
- **Platzi** Linear Algebra for ML, Math Functions for DS & AI, Math Language & Notation, Math Fundamentals

LANGUAGES

Spanish
native

English
certified intermediate (2024); advanced in preparation